

Mini-Course on Generative AI Applications

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Session 2

- ▶ Prompt Engineering
- ▶ Practical Examples

Nov 4

What is Prompt Engineering?

The art of creating effective instructions/prompts to get the best results from AI models.

Examples

- ▶ Zero-shot
- ▶ Few-shot
- ▶ Chain-of-Thought (CoT)

Zero-shot

Definition

Zero-shot is the ability of a model to perform a task it was not explicitly trained on, without any examples.

Example

Input: "The weather is nice today."

Output: "Le temps est agréable aujourd'hui."

Definition

Few-shot gives the model some example to learn and perform a task.

Example

Examples:

- ▶ "I loved the movie, it was fantastic!" → Positive
- ▶ "The film was dull and boring." → Negative

Input: "An excellent performance by the lead actor."

Output: Positive

Chain-of-Thought (CoT)

Definition

Chain-of-Thought prompting guides the model to reason through problems step-by-step.

Example

Input: "If you have 3 apples and you buy 2 more, how many apples do you have in total?"

Chain-of-Thought:

1. Start with 3 apples.
2. Buy 2 more apples.
3. Total apples = $3 + 2 = 5$.

Output: You have 5 apples in total.

Prompt Engineering - Some Resources

- ▶ <https://arxiv.org/pdf/2406.06608>
- ▶ <https://github.com/claudiobottari/prompteng>
- ▶ <https://www.promptingguide.ai/>
- ▶ <https://github.com/aws-samples/claude-prompt-generator/tree/main>
- ▶ <https://github.com/dair-ai/Prompt-Engineering-Guide>
- ▶ Many Youtube Videos

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Best Strategy?

Testing and Learning by Doing

Prompting Techniques 1

Task/Instructions: Classify as positive or negative: `TEXT`

Output Formatting: Summarize `TEXT` into a CSV.

Style Instructions: Write a `clear paragraph using simple words` explaining how `remote work changes the economy`.

Prompting Techniques 2

Additional Information/Context: Consider factors such as inflation, cost of living, and labor market dynamics when analyzing the impact of minimum wage increases on employment rates.

Role/Persona: As an urban economist analyze how remote work reshapes city geography.

Plan-and-Solve Prompting: Let's first understand the problem and devise a plan to analyze how minimum wage increases have affected employment rates in the U.S. over the past ten years.

Prompting Techniques 3

Emotion Prompting: This question very important for me.

Rephrase and Respond (RaR): Before analyzing, please rephrase the question about

Self Prompting: Rewrite the prompt to focus solely on the question, removing any unrelated information: "How have minimum wage increases affected employment rates in the U.S. over the past decade?" Then, provide an analysis.

Prompting Techniques 4

Prompt Mining: Process of discovering optimal "words" in prompts

Re-reading: Read the question again and then answer.

Self-Ask: What additional information is needed to analyze **TEXT**?

Prompting Techniques 5

Rag prompting 1: Read the PDF and

Rag prompting 2: Search the web and

Multimodal prompting: Read this CSV file and give some trend analysis like the attached plot.

Prompting Techniques 6

Multiple Perspectives: Analyze from the view of `investors, businesses, government, residents`

Break Down Complex Tasks: Analysis components: `markets, business, finance, labor, infrastructure`

Confirm Understanding

Prompting Techniques 7

Set Knowledge Cutoff Dates: Use **2019-2023 data** for

Ask for Examples: Provide **examples** of

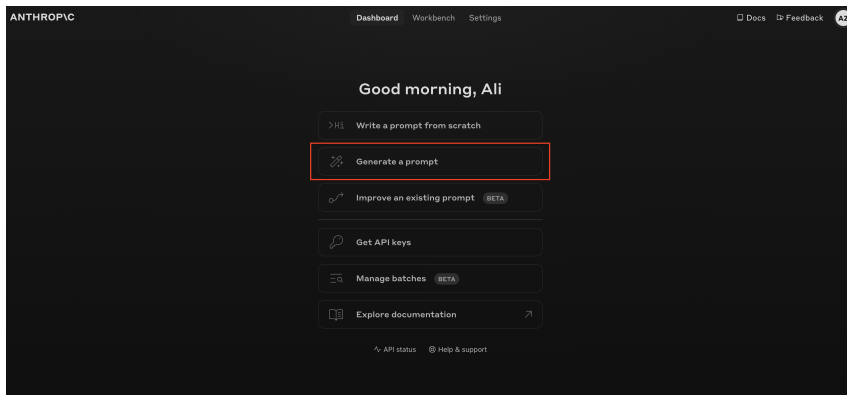
Set Length Constraints: In **500 words**, analyze

Some Tips

1. **Blending**
2. **Reset**

Prompt - When is very important to you

- ▶ Prompt Databases
- ▶ Prompt Generators



<https://console.anthropic.com/dashboard>

Thank you!